Concavity and inflection points

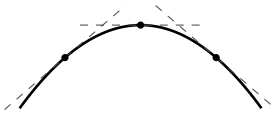
Friday, November 6, 2026 *Transcribed by Cleo McBride*

How do the slopes of the tangent lines change as we move from left to right?

Definition 1 (Concavity). On an interval where f is differentiable, its graph is *concave up* if f' is increasing, and *concave down* if f' is decreasing.



Increasing slopes



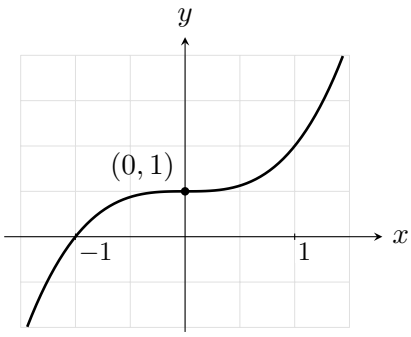
Decreasing slopes

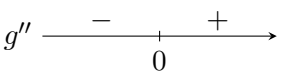
Theorem 1 (Concavity test). If $f'' > 0$ throughout an interval, the graph of f is concave up there. If $f'' < 0$ throughout an interval, it is concave down. This follows by applying the increasing and decreasing test to f'.

Definition 2 (Inflection point). If f is continuous at c and the graph changes concavity at c, then $(c, f(c))$ is an *inflection point*.

Example 1. Where does the graph of $g(x) = x^3 + 1$ change concavity?

$$g'(x) = 3x^2, \qquad g''(x) = 6x.$$





The graph is concave down on $(-\infty, 0)$ and concave up on $(0, \infty)$. Thus $(0, 1)$ is an inflection point. Although the tangent there is horizontal, g increases through it.

The second derivative test

At a horizontal tangent, can we classify a critical number using just $f''(c)$?

Theorem 2 (Second derivative test). Suppose $f'(c) = 0$ and f'' is continuous on an open interval containing c.

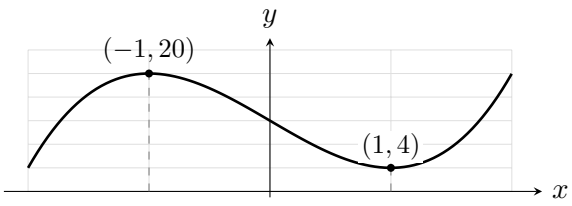
$$\begin{aligned} f''(c) > 0 &\implies f(c) \text{ is a strict local minimum,} \\ f''(c) < 0 &\implies f(c) \text{ is a strict local maximum.} \end{aligned}$$

When $f''(c) = 0$, the test is inconclusive.

Proof. If $f''(c) > 0$, continuity keeps f'' positive near c. Thus f' increases through $f'(c) = 0$, from negative values to positive values. The first derivative test gives a local minimum. When $f''(c) < 0$, f' decreases through zero, giving a local maximum. □

Example 2. Classify the critical numbers of $f(x) = 4x^3 - 12x + 12$. Since $f'(x) = 12(x-1)(x+1)$, the critical numbers are -1 and 1. Using $f''(x) = 24x$ gives

$$\begin{aligned} f''(-1) = -24 < 0 &\implies \text{local maximum at } (-1, 20), \\ f''(1) = 24 > 0 &\implies \text{local minimum at } (1, 4). \end{aligned}$$



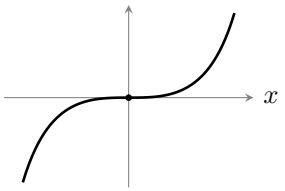
Example 3. What does the second derivative test tell us about $h(x) = x^2 \tan x$ at $x = 0$?

$$\begin{aligned} h'(x) &= 2x \tan x + x^2 \sec^2 x, \\ h''(x) &= 2 \tan x + 4x \sec^2 x + 2x^2 \sec^2 x \tan x. \end{aligned}$$

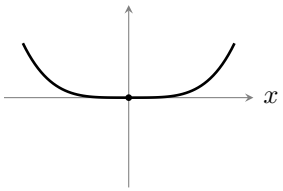
Substituting $x = 0$ gives $h'(0) = 0$ and $h''(0) = 0$. The test is inconclusive.

For $0 < |x| < \pi/2$, $x^2 > 0$ and $\tan x$ has the same sign as x. Thus $h(x) < h(0)$ just to the left of zero and $h(x) > h(0)$ just to the right. There is no local extremum at 0.

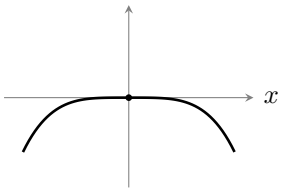
Remark. The same test result can occur at a local minimum (x^4 at 0) or a local maximum ($-x^4$ at 0). In both cases the graph keeps the same concavity across zero.



$y = x^2 \tan x$



$y = x^4$



$y = -x^4$